Ch 6.3, 6.4: PLS; Curse of Dimensionality

Lecture 21 - CMSE 381

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Announcements

Last time:

- PCA/PCR

This lecture:

- Finish PCR
- 6.3: PLS
- 6.4: Issues with higher dimensions

Announcements:

- Homework #5 due Friday

Section 1

Last time

Shrinkage

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

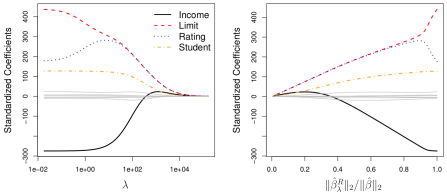
Least Squares:
No constraints

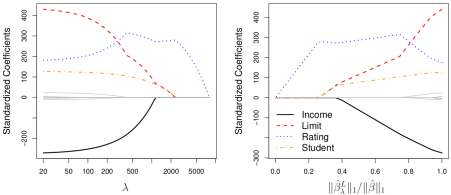
Ridge:

$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$





Also, find best choice of λ or s using CV

Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

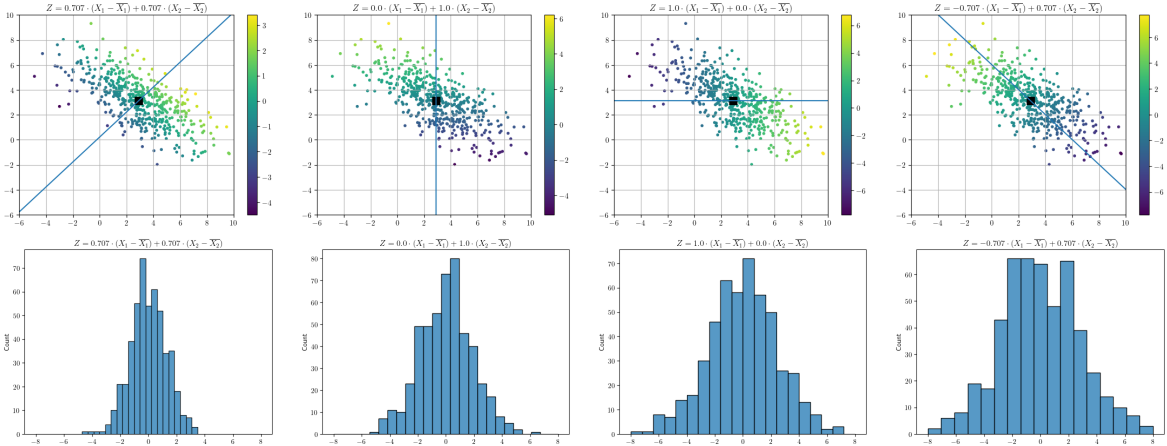
The goal:

- Find good φ's (PCA)
- Fit regression model on Z_i's using least squares (PLS)

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

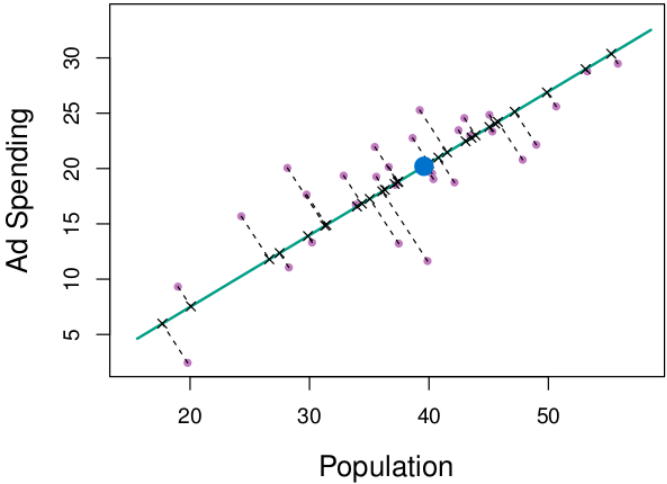
- Hope that lower dimensions means less overfitting

PCA - First PC



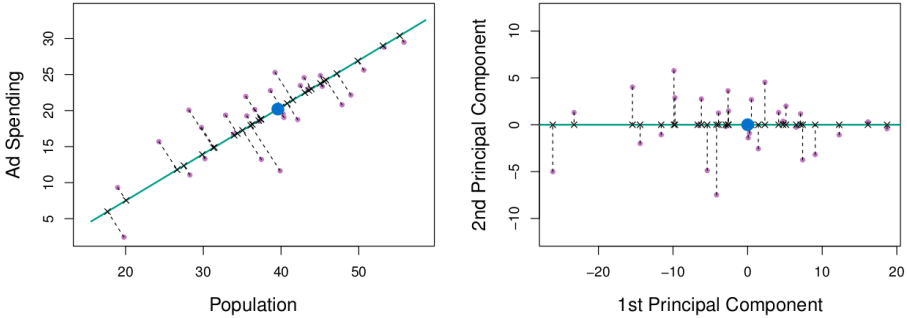
higher variance of projection on the further right pic

Projection onto first PC



$$Z_1 = 0.839 \cdot (\texttt{pop} - \overline{\texttt{pop}}) + 0.544 \cdot (\texttt{ad} - \overline{\texttt{ad}})$$

Drawing points in PC space



Interpretation of PCR coefficients

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

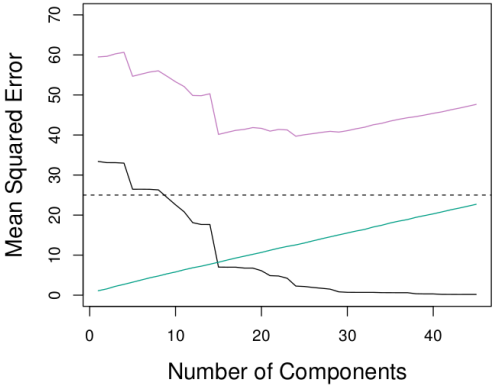
Learned model:

$$y = \theta_0 + \theta_1 Z_1 + \cdots + \theta_M Z_M$$

- Do example showing that coefficients of Z_i can't be used to say anything about X_i's
- $p = 3, M = 2$
- $\varphi = \begin{pmatrix} 1 & -1 \\ 3 & 1/2 \\ 0 & 1/2 \end{pmatrix}$
- Learned model $y = 2Z_1 - Z_2$
- $y = 2(X_1 + 3X_2 + 0X_3) - (-X_1 + \frac{1}{2}X_2 + \frac{1}{2}X_3)$
- $= 3X_1 + 2.5X_2 - 0.5X_3$
- Conclusion: can't directly use coefficients learned to interpret original inputs!
- Each PC is a linear combination of the originals, so can go backwards to figure out whaat the coefficients would have been, see left figure

Bias variance trade-off

Example with simulated data: $n = 50$ observations of $p = 45$ predictors
Y is a function of **all predictors**

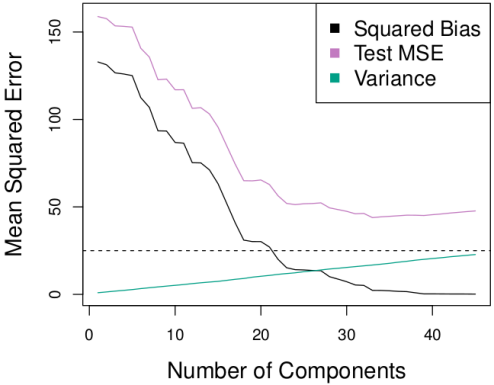


- Right side is usual least squares estimate
- Even better improvement using PCR than previous
- note the chnage in y-axis values

Bias variance trade-off

Example with simulated data: $n = 50$ observations of $p = 45$ predictors
Y is a function of **2 predictors**

- Right side is usual least squares estimate
- Typical U shape for Test MSE
- PCR with good choice of number of components improvement over plain least squares



Properties of PCR

- Note better in previous example because lots of PCs needed
- It does better when info contained in the first few PCs
- Number of PCs to use can be picked with CV
- Standardization needed to make sure big data values don't skew the results
- Not a feature selection method because we don't get a subset of features, we get a collection of linear combos of features

Section 2

Partial Least Squares (PLS)

Supervised alternative

PCR: Non-supervised

- No input from the Y values before learning the PCs.
- No guarantee that directions that explain Xs help to predict the Ys

Partial Least Squares (PLS):

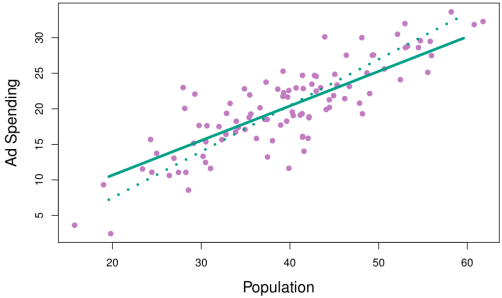
- Identify new features $Z_1, \cdots, Z_M$ linear combos of original where quality measure involves Y
- Fit linear model using least squares on these M features

First direction Z_1 for Partial Least Squares (PLS)

- Set φ_{j1} equal to the coefficient from simple linear regression of Y onto X_j
  This means we do p linear regressions, not that we take the multivariable linear regression coefficients
- The first direction is

$$Z_1 = \sum_{j=1}^{p} \varphi_{j1} X_j$$

- PLS places highest weight on variables most strongly related to the response
- In example, PLS chooses direction less change in ad direction than pop relative to PCA



Ex. Prediction of Y =Sales (not shown) on X_1 =Population and X_2 =Ad Spending

- Solid green: First PLS direction
- Dashed: First PC direction

Second (and more) PLS directions

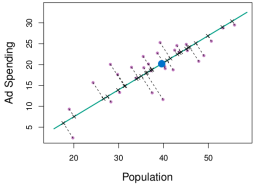
- Regress each variable on Z_1 and take residuals *This is the remaining info not explained by first PLS direction*
- Compute Z_2 using *orthogonalized data* same as for Z_1
- Number M can be picked using CV after standardizing predictors
- Mixed bag in terms of results..... in practice often performs no better than ridge or PCR

Code example on hitters data

PCR vs PLS

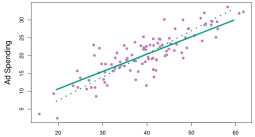
PCR

- Unsupervised dimensionality reduction + linear regression
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



PLS

- Supervised dimensionality reduction
- Choose component Z_1 by using simple regression coefficients of each X_i onto Y
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained
- Not a particular benefit, so usually default to PCA unless you have a good reason for this



Section 3

Issues in Higher Dimensions

Goal of this chapter

Classes of methods

- Subset selection: identify subset of predictors, fit model using least squares on smaller set of variables
- Shrinkage:
  - Estimated coeff are shrunken towards zero relative to the least squares estimate
- Dimension reduction: Project the p-dimensional data into M-dimensional subspace, $M < p$. Likely after the midterm.

High-Dimensional Data

Low-Dimensions

$n \gg p$

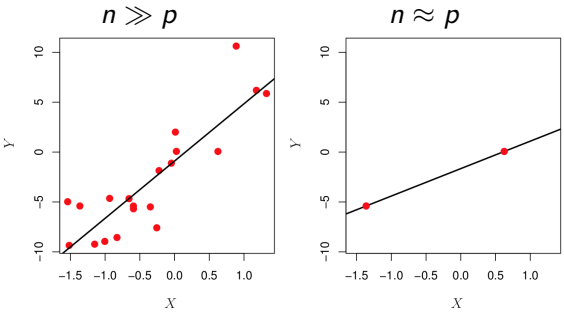
- Low here means p is low, or at least small relative to n
- Can do all the stuff we've talked about so far
- Historically, this is the data we had to work with
- e.g. Predict blood pressure on age, sex, BMI. $p = 3$ and we have hundreds of thousands of patients to work with

High-Dimensions

$n \ll p$

- New technology makes this possible
- Predict blood pressure based on single nucleotide polymorphisms (SNPs, individual DNA mutations). $n \approx 200$, $p \approx 500,000$
- Search engine bag of words model. Maybe you can access a few hundred people, but each word is scored as present (0) or (1) absent. Huge number of words
- Classical approaches not appropriate since overfitting, bias-variance tradeoff
- Issues show up even if p is slightly smaller than n

What goes wrong?

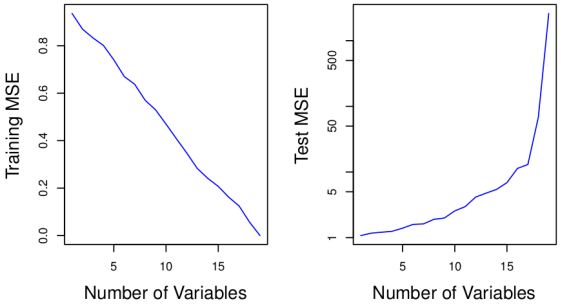


- $p = 1$ feature
- $n \gg p$
- Not perfect fit, but more reasonable fit

- $p = 1$ feature
- $n = 2$ just barely bigger than p
- Fit will always be perfect

- If this is 2 samples from the same as the left, result is a pretty bad model
- Idea: if $p > n$ or $p \approx n$, then line is too flexible so overfits

More issues with least squares on big p



- $n = 20$
- Regression on $p = 1, \cdots, 20$
- Y completely unrelated to variables

- Training MSE goes to 0 even though this whole set is garbage
- Test MSE gets worse because additional predictors lead to increase in variance of coeff estimates

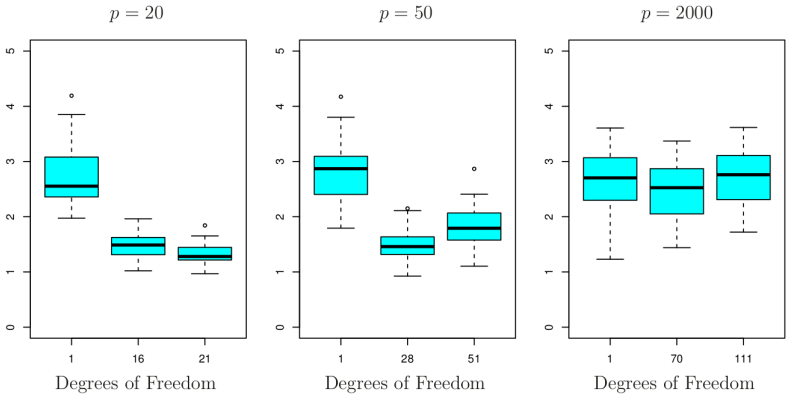
- If you just look at R^2 or training MSE, you would suspect that more variables is better
- Extra care needed when analyzing data with very large p
- Like R^2, C_p, AIC, BIC don't work well in high dimensions. Cause is $\hat{\sigma}^2$ is problematic.

The answer to dealing with big p

Be less flexible!

- Forward step selection
- ridge regression
- lasso
- principal components regression

Example with Lasso



- $n = 100$
- Boxplots = Test MSE
- DF = # non-zero coeffs *Messed with λ then reported this*

- High λ means fewer non-zero coeffs means high λ is to the left

- Small p: low val. set error with λ small
- p larger means larger λ used

Key points

- (1) regularization or shrinkage plays a key role in high-dimensional problems,
- (2) appropriate tuning parameter selection is crucial for good predictive performance, and
- (3) the test error tends to increase as the dimensionality of the problem (i.e. the number of features or predictors) increases, unless the additional features are truly associated with the response.

Curse of dimensionality

Phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings.

- You'd think more variables means better model fit
- IN previous example, test MSE doubles as p increases from 20 to 2,000.
- Addition additional features that are TRULY associated with response will improve fitting model
- Adding noise features will result in deterioration
- Increase risk of overfitting without upside in test set error

Interpretation in high dimensions

Multi-collinearity: the concept that the variables in a regression might be correlated with each other

- High dim: any variable in the model can be written as a linear combination of all of the other variables in the model.

- can never know exactly which variables (if any) truly are predictive of the outcome,
- can never identify the best coefficients for use in the regression.
- At most, we can hope to assign large regression coefficients to variables that are correlated with the variables that truly are predictive of the outcome.
- Resulting model is *one of many possible*

Reporting errors in high dimensions

- When $p > n$, easy to obtain a useless model that has zero residuals.
- Don't use sum of squared errors, p-values, R 2 statistics, or other traditional measures of model fit on the training data as evidence of a good model fit
- Reporting this fact might mislead others into thinking that a statistically valid and useful model has been obtained
- in fact this provides absolutely no evidence of a compelling model.

- Report results on an independent test set, or cross-validation errors.
- For instance, the MSE or R 2 on an independent test set is a valid measure of model fit, but the MSE on the training set certainly is not.

Next time

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Subset selection	6.1	HW #4 Due Weds 10/9
17	Fri	10/11	Shrinkage: Ridge	6.2.1	
18	Mon	10/14	Shrinkage: Lasso	6.2.2	
19	Wed	10/16	Dimension Reduction	6.3	
20	Fri	10/18	Overflow, Possibly more dimension reduction?		HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		